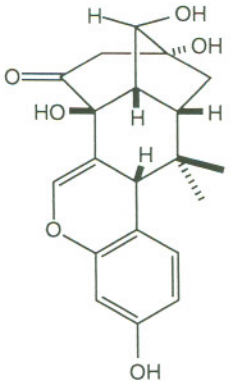
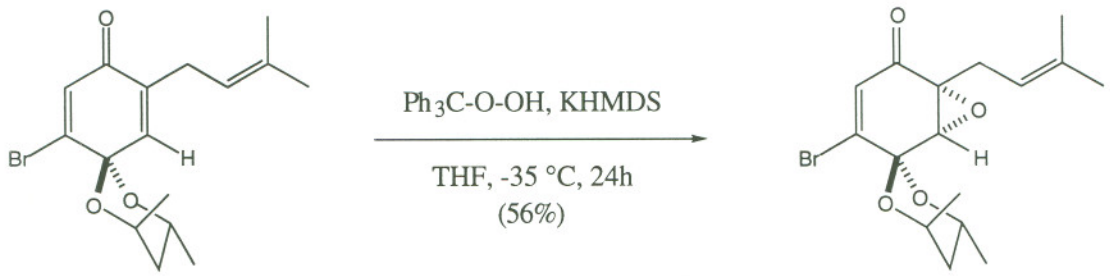
"Enantioselective Total Synthesis of Miroestol"
E. J. Corey and L. I. Wu
J. Am. Chem. Soc. **1993**, *115*, 9327.



Miroestrol **1**

In the highly efficient convergent total synthesis of miroestrol (**1**) (referred to as a "fountain of youth" in traditional folk medicine) by Corey and Wu, a number of selective transformations were performed which required very specific conditions in order to accomplish.

1) Conversion of **2** to **3** required the chiral ketal derivative in order to accomplish the desired epoxidation with the stereoselectivity shown.



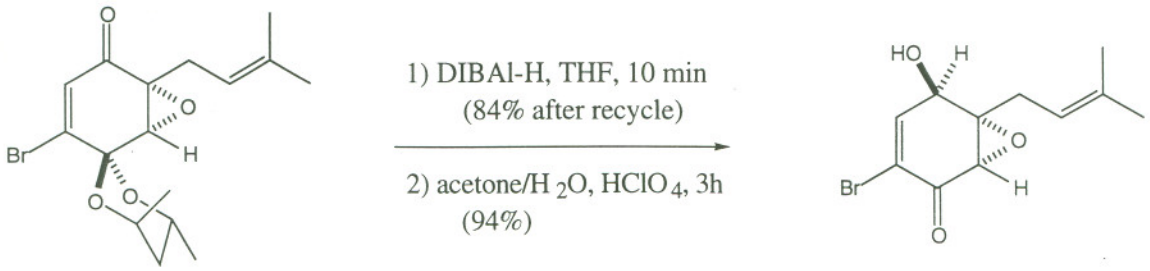
2 [KHMDS = $K^+(Me_3Si)_2N^-$] **3**

(a) Draw the other possible conformation of chiral ketal quinone **2** and explain why **2** is prefered over the other possible conformer.

(b) Rationalize the chemoselectivity in the reaction.

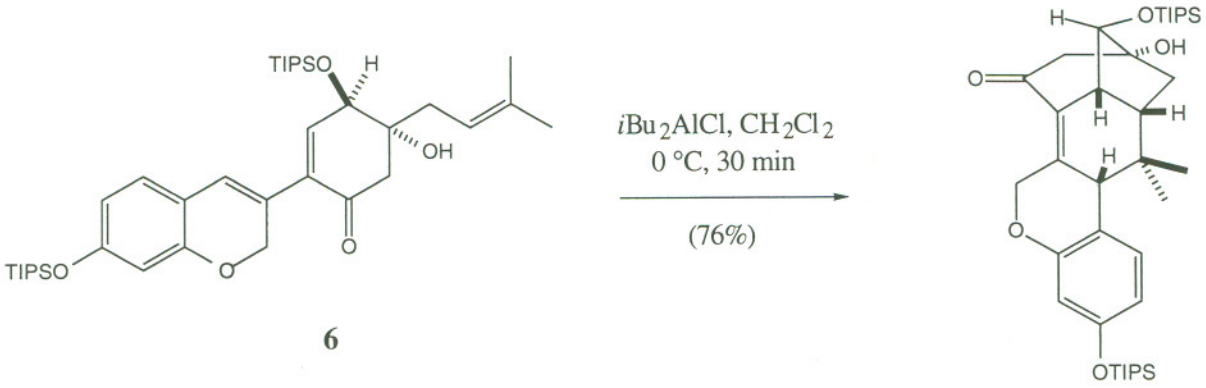
(c) Describe at least one unique quality of the peroxide used in this reaction that would dictate its use over other possible alkyl peroxides one could choose.

2) Rationalize the stereoselectivity observed in the reduction involved in conversion of **3** to **4**.



3 [DIBAl-H = $(iBu)_2Al$-H] **4**

3) Provide a complete detailed mechanism of the Lewis acid catalyzed cyclization of **6** to **7** showing all intermediates involved and arrows which **clearly** show the electron flow and charges as necessary.



6 **7**